

Where Does the “Free Lunch” Come From?

Disentangling Outer Momentum from Worker Diversity in Low-Communication LLM Training

Detailed project plan for a reproducible paper submission

Core idea: Run a controlled, compute- and token-matched head-to-head among synchronous data-parallel AdamW, multi-worker DiLoCo, single-worker SNOO, and outer-optimizer ablations. The paper asks whether reported gains come from low-communication worker divergence, from the Nesterov outer lookahead mechanism, or from an interaction between the two.

Primary reference	Charles et al. (2025), Scaling Laws for DiLoCo, arXiv:2503.09799
Key confound identified	Single-worker DiLoCo/SNOO can itself improve optimization, so distributed gains must be separated from outer-optimizer gains.
Recommended scope	60M-410M parameter main study, with a 1B confirmation run if compute allows.
Primary outcome	Validation loss at matched tokens/FLOPs, plus communication volume and downstream accuracy.
Submission posture	Reproduction + causal decomposition + robust negative/positive result, not a simple benchmark rerun.

1. Executive Summary

The strongest paper opportunity is a controlled causal decomposition of the “free lunch” claim in low-communication language-model training. Scaling Laws for DiLoCo reports that well-tuned DiLoCo can match or exceed synchronous data-parallel training under fixed compute while communicating far less. Later work on SNOO shows that a single-worker version of the same two-loop construction can itself outperform AdamW, suggesting that part of the apparent distributed advantage may come from the outer Nesterov update rather than from low-communication distribution.

This project therefore treats worker distribution and outer lookahead as separate experimental factors. The central study compares synchronous data-parallel AdamW, single-worker AdamW, single-worker SNOO, multi-worker periodic averaging without outer momentum, and multi-worker DiLoCo under matched token budgets, model FLOPs, learning-rate search effort, data order, and evaluation protocol. A second matching protocol controls effective/global batch size to expose whether any advantage is simply a gradient-noise effect.

Novelty candidate: A publishable contribution is not “DiLoCo versus AdamW” by itself. The candidate novelty is the factorial attribution: quantify how much of any quality gain is explained by (A) pseudo-gradient Nesterov, (B) independent worker trajectories / local drift, and (C) their interaction, under equalized compute and tokens. The literature scan did not reveal an existing paper that cleanly performs this exact decomposition across model scales. Novelty should still be re-checked immediately before submission.

2. Proposed Paper Title and One-Sentence Claim

Recommended title

Where Does the Free Lunch Come From? Disentangling Outer Momentum from Worker Diversity in Low-Communication LLM Training

Alternative titles

- The Free Lunch, Decomposed: A Controlled Study of DiLoCo, SNOO, and Data-Parallel AdamW
- Is Low Communication the Advantage? Causal Ablations of DiLoCo and Single-Worker Outer Momentum
- Distributed Gain or Optimizer Gain? Reproducing DiLoCo Under Matched Tokens, FLOPs, and Gradient Noise

Target claim

At fixed training tokens and comparable model FLOPs, the apparent advantage of low-communication training can be decomposed into an outer-optimizer component and a multi-worker trajectory component. The paper will report the size, stability, and scale-dependence of each component rather than assuming either mechanism is responsible.

3. Why This Is Timely: Literature Positioning

Work	What it establishes	Implication for this paper
DiLoCo (2023/2024)	Introduced infrequent worker synchronization with AdamW inner optimization and Nesterov outer optimization; reported comparable quality with much less communication.	Establishes the algorithm and single-worker acceleration observation.
Streaming DiLoCo (2025)	Reduced peak bandwidth via fragmented, overlapped, quantized synchronization; framed the goal as a distributed “free lunch.”	Makes the free-lunch claim practically important, but its system contribution does not isolate optimizer causality.
Scaling Laws for DiLoCo (2025)	Studied scale, replica count, token budget, and hyperparameters; reported predictable scaling and cases outperforming data parallel.	Primary result to reproduce and stress-test at the project’s scale.

Work	What it establishes	Implication for this paper
SNOO (2025)	Argued that applying Nesterov momentum to pseudo-gradients is highly effective even in one-worker training.	Creates the key confound: an apparent distributed gain can be an optimizer gain.
GPA / smoothing work (2025-2026)	Improves or analyzes non-distributed two-loop/averaging optimizers.	Shows the single-worker mechanism is an active optimization research direction.
High-dimensional theory / momentum restart (2026)	Analyzes two-phase optimizers and outer momentum behavior theoretically.	Provides theory-facing diagnostics and supports a mechanistic discussion.

What is still missing

- A direct head-to-head in one codebase where DDP AdamW, SNOO, FedAvg/local-SGD style periodic averaging, and DiLoCo share the same model, tokenizer, data pipeline, schedule family, token budget, and evaluation harness.
- A factorial attribution that isolates the outer Nesterov effect from the effect of independent local trajectories and their averaging.
- Two fairness views: token/FLOP matching and global-batch/gradient-noise matching. These answer different questions and prevent a misleading “free” gain caused by batch-size changes.
- Confidence intervals across seeds and model scales, with a predeclared primary endpoint rather than selecting the best-looking checkpoint or downstream task after training.

4. Research Questions and Pre-Registered Hypotheses

ID	Question
RQ1	Can the reported DiLoCo quality advantage over synchronous data-parallel AdamW be reproduced at our scale when tokens and training FLOPs are controlled?
RQ2	How much of the gap is reproduced by single-worker SNOO, where there is no cross-worker distribution at all?
RQ3	After removing outer Nesterov, does independent local training plus periodic averaging still improve quality relative to DDP?
RQ4	Does worker diversity interact positively or negatively with the outer optimizer as model size, sync interval H, or number of workers M changes?
RQ5	Are conclusions preserved when effective/global batch size is controlled separately from total-token and compute matching?
RQ6	What is the Pareto frontier among validation quality, training compute, bytes communicated, and wall-clock time under bandwidth constraints?

Hypotheses

- H1: SNOO will recover a substantial fraction of any DiLoCo-versus-DDP loss improvement, implying that outer pseudo-gradient momentum is a major contributor.
- H2: Multi-worker local drift without outer momentum will not consistently reproduce the full gain and may become harmful at large H unless hyperparameters are retuned.
- H3: The interaction term between worker diversity and outer Nesterov will vary with model scale and H; a single global claim of “distributed optimization is better” will be too coarse.
- H4: Some apparent gains will shrink under global-batch matching, revealing a gradient-noise/batch-size contribution.

- H5: DiLoCo will retain a communication advantage even in regimes where its quality advantage disappears, so the paper can remain useful if the strongest quality claim is not reproduced.

5. Core Experimental Design

5.1 Methods to compare

Method	Workers	Sync	Outer update	Purpose
DP-AdamW	M synchronized workers	Every step	None beyond AdamW	Gold-standard synchronous data-parallel baseline.
SW-AdamW	1 worker	None	None	Single-worker control; helps separate data parallelism from optimizer behavior.
SNOO	1 worker	Local outer update every H steps	Nesterov on pseudo-gradient	Isolates the outer lookahead mechanism with zero worker diversity.
PeriodicAvg	M independent workers	Average every H steps	Identity / plain averaging	Isolates local drift + averaging without Nesterov outer momentum.
DiLoCo	M independent workers	Average every H steps	Nesterov on averaged pseudo-gradient	Full treatment.
Clone-DiLoCo	M workers fed identical batches/order	Average every H steps	Nesterov	Controls distributed machinery while suppressing data/gradient diversity.
Shuffle-Control	M workers, independently shuffled from same corpus	Average every H steps	Nesterov	Measures ordinary stochastic worker diversity without non-IID partitioning.

Minimum publishable comparison: DP-AdamW, SNOO, PeriodicAvg, and DiLoCo are the minimum four conditions. Clone-DiLoCo is highly recommended because it is the cleanest “same mechanism, no useful worker diversity” control.

5.2 Two fairness protocols

Protocol A: token/FLOP matched (primary)

- All methods process the same total number of training tokens per model scale.
- Report estimated training FLOPs using the same accounting rule; include any extra outer-update compute, even if small.
- Use the same tokenizer, sequence length, model architecture, weight decay convention, warmup fraction, evaluation frequency, and data sampling universe.
- This protocol answers: given the same amount of model-training work, which algorithm produces the best model and at what communication cost?

Protocol B: effective/global-batch matched (secondary)

- Match tokens contributing to one comparable update timescale. For DDP this is the all-reduced global batch; for low-communication methods explicitly define the window over which pseudo-gradients are formed.
- Use gradient accumulation where needed so that SNOO and DDP can be compared at a similar effective batch without changing total tokens.

- This protocol answers: are observed improvements mostly a consequence of changed gradient noise or larger effective batch?

5.3 Recommended model and token scales

Stage	Model size	Token budget	Role	Seeds
Pilot	~35-60M	0.5-1.0B	Debug correctness, tune search ranges, estimate variance.	3 seeds for key cells
Small	~100-160M	2-4B	Main factorial sweep over H and outer hyperparameters.	3 seeds
Medium	~300-410M	6-10B	Confirm effect scaling with realistic cost.	3 seeds core, 1 seed wide sweep
Large confirmation	~1B	20-25B or budget-limited matched tokens	One narrow confirmation of the winning/diagnostic conditions.	1-2 seeds

If compute is limited, prioritize more seeds at 100-410M over a single spectacular 1B run. The paper’s contribution is causal attribution and reproducibility; variance estimates matter more than scale theater.

5.4 Data

- Reproduction corpus: use C4 or the exact preprocessing closest to the Scaling Laws for DiLoCo setup when practical, so the contested claim is tested in-distribution with prior work.
- External-validity corpus: use one modern web corpus such as Dolma or a fixed FineWeb/FineWeb-Edu slice, with a frozen manifest and hash. Do not mix corpora between methods.
- For worker-distribution ablations, create explicit partition modes: IID random shards, domain-skewed shards, and clone/identical-stream control. Record shard entropy and token counts.
- Never let different methods see different total tokens because one run exhausted or repeated a shard earlier. Sampling should be deterministic from a shared global example index where possible.

6. Required Experiments

6.1 Experiment E0 - Training harness equivalence

Goal: prove that the framework does not introduce accidental differences before testing the research claim.

- Run DP-AdamW with 1, 2, and M devices while holding global batch and seed logic fixed. Confirm loss traces agree within expected floating-point/non-deterministic tolerance.
- Verify token counters, LR schedule step units, weight decay, gradient clipping, mixed precision, optimizer epsilon/betas, and checkpoint resume behavior.
- Measure actual tokens/s, model FLOPs utilization, peak memory, and communication bytes. This produces the systems baseline used later.

6.2 Experiment E1 - Direct reproduction of the “free lunch”

- At 100-160M and 300-410M, compare DP-AdamW versus DiLoCo at M = 4 or 8 and H in {30, 100, 300 or nearest feasible values}.
- Tune inner learning rate for each method using the same search budget. Tune DiLoCo outer LR/momentum on a declared grid, then freeze the policy before medium-scale confirmation.
- Primary endpoint: validation cross-entropy at a fixed token budget. Secondary: downstream zero-shot accuracy, communication bytes, and wall time under measured and simulated bandwidth.

6.3 Experiment E2 - Outer optimizer attribution

- Compare DP-AdamW, SW-AdamW, SNOO, PeriodicAvg, and DiLoCo at identical model size and token budget.
- Add DiLoCo with outer momentum = 0 and/or outer LR chosen so the step reduces to simple averaging. This provides a smooth bridge rather than only categorical methods.
- Compute the “SNOO recovery fraction”: $(\text{Loss_DP} - \text{Loss_SNOO}) / (\text{Loss_DP} - \text{Loss_DiLoCo})$, with uncertainty from seeds. Interpret only when the denominator is meaningfully non-zero.

6.4 Experiment E3 - Worker diversity attribution

- Compare Clone-DiLoCo, IID-shard DiLoCo, and domain-skewed DiLoCo with identical outer hyperparameters.
- Log pairwise cosine similarity of worker pseudo-gradients, norm dispersion, model-distance between workers, and the angle between the averaged pseudo-gradient and a synchronous reference gradient sampled at sync points.
- Test whether quality correlates with moderate diversity, extreme drift, or neither. Avoid claiming causality from correlation unless the controlled clone/IID/skew intervention supports it.

6.5 Experiment E4 - Batch/noise control

- Repeat the core four-method comparison under effective/global-batch matching. Use accumulation or microbatch changes without altering total tokens.
- Estimate gradient-noise scale or a lightweight proxy (e.g., gradient norm variance across microbatches) at selected checkpoints.
- If the DiLoCo gap collapses here, the paper’s contribution becomes an explanation: the free lunch was partly a batch/noise change rather than a uniquely distributed optimization effect.

6.6 Experiment E5 - Scale and H interaction

- Run at least three model sizes and 3-4 sync intervals H. Fit simple response surfaces or log-linear scaling relations for loss gap versus model size, H, and M.
- Do not fit a grand scaling law unless the number of scales supports it. A robust interaction plot with uncertainty is preferable to overclaiming a power law.
- Key plot: delta validation loss relative to tuned DP-AdamW versus model scale, with separate curves for SNOO, PeriodicAvg, and DiLoCo.

6.7 Experiment E6 - Communication-quality Pareto frontier

- Measure payload bytes from the training implementation rather than only theoretical parameter counts.
- Report communication per processed token and total TB transferred. Include peak bandwidth requirement if burst synchronization matters.
- Replay measured compute traces through bandwidth caps (for example 1, 10, 25, 100 Gbit/s) to estimate wall-clock utilization. Clearly label simulation versus measured wall time.

6.8 Experiment E7 - Robustness checks

- Seeds: at least 3 for primary 100-410M cells. Report mean, standard deviation, and paired differences when data order can be aligned.
- Outer hyperparameter sensitivity: heatmap over outer LR and momentum for one small and one medium model.
- Learning-rate fairness: equal number of tuning trials for each algorithm family; report the search space and selection rule.
- Checkpoint selection: primary result uses the fixed final token budget, not best validation checkpoint. Best-checkpoint numbers may be secondary.
- Dataset replication: repeat the central comparison on a second corpus for one model scale.

7. Concrete Experimental Matrix

ID	Scale	Methods	H	Seeds	Primary purpose
E0	60M	DP-AdamW 1/2/4 device	H=1	1	Harness parity; tokens/s; comm bytes
E1	160M	DP-AdamW, DiLoCo	30/100/300	3	Reproduce quality gap
E2	160M	DP, SW-AdamW, SNOO, PeriodicAvg, DiLoCo	100	3	Main causal decomposition
E2b	410M	DP, SNOO, PeriodicAvg, DiLoCo	100	3	Scale confirmation
E3	160M	Clone/IID/skew DiLoCo	100	3	Worker diversity mechanism
E4	160M	DP, SNOO, PeriodicAvg, DiLoCo	100	3	Global-batch matched
E5	60M/160M/410M	DP, SNOO, DiLoCo	30/100/300	1 wide + 3 key	Scale x H interaction
E6	160M/410M	DP, DiLoCo	best H	n/a	Communication Pareto
E7	160M	core methods	best H	3	Second corpus + sensitivity
E8 optional	1B	DP, SNOO, DiLoCo	best H	1-2	Large-scale confirmation

Estimated run count strategy

Use staged pruning. First complete E0 and a 60M pilot. Next perform broad single-seed sweeps at 160M to locate viable H/outer-LR regions. Only then spend three seeds on the predeclared primary cells. Medium and 1B runs should confirm conclusions, not discover hyperparameters from scratch.

8. Metrics and Statistical Analysis

Class	Metric	Why it matters
Primary	Validation cross-entropy / loss at exactly matched token budget	Direct training-quality endpoint; least cherry-pickable.
Primary systems	Total bytes communicated and communication bytes per token	Quantifies the low-communication benefit.
Secondary	Perplexity (derived from loss)	Reader-friendly language-model metric.
Secondary	HellaSwag, PIQA, ARC-Easy or a small fixed zero-shot suite	Checks whether loss differences transfer to capability metrics.
Efficiency	Tokens/s, step time, MFU, wall-clock under bandwidth cap	Distinguishes algorithmic quality from infrastructure benefit.
Mechanistic	Pseudo-gradient cosine similarity, norm variance, worker parameter distance	Tests the worker-diversity story.
Optimizer	Outer momentum-buffer norm, pseudo-gradient/velocity angle	Tests whether outer memory drives acceleration.
Stability	Seed std, failure/divergence count, sensitivity surface	Prevents one lucky run from defining the paper.

Statistical plan

- Predeclare the primary comparison at one model scale and H before the final multi-seed runs.

- For each primary method, report mean +/- standard deviation across seeds and a 95% bootstrap confidence interval for the pairwise loss difference.
- When runs share the same initialization/data order, use paired seed-level differences as the main uncertainty estimate; otherwise use unpaired bootstrap intervals.
- Do not convert tiny loss differences into strong language unless they are stable across seeds and at least one second scale or corpus.
- Report all failed or diverged hyperparameter trials in an appendix table so stability is part of the evidence.

9. Implementation Requirements

9.1 Reproducibility controls

- One repository and training entry point for every method; algorithm differences selected by configuration rather than separate codebases.
- Immutable config files committed for every reported run, with Git commit hash stored in checkpoints/logs.
- Deterministic data-shard manifest with corpus version, tokenizer version, seed, and total token count.
- Unified token counter that drives LR schedules and evaluation triggers. Avoid using “optimizer step” as the universal clock because H and worker count change its meaning.
- Same mixed-precision policy, gradient clipping, dropout, initialization, sequence length, and optimizer implementation wherever conceptually applicable.
- Checkpoint optimizer state for both inner and outer optimizers; verify resume equivalence in E0.
- Log actual communication payload size and synchronization duration at every outer step.
- Export a compact run summary JSON/CSV for paper tables, rather than manually copying values from dashboards.

9.2 Suggested configuration schema

Group	Fields
model	n_layers, d_model, n_heads, vocab_size, seq_len, parameter_count
data	corpus, split, tokenizer, shard_mode, shard_seed, worker_shards, token_budget
inner optimizer	AdamW lr, betas, eps, weight_decay, warmup_tokens, decay schedule, grad_clip
outer optimizer	type, outer_lr, momentum, Nesterov flag, reset behavior
distribution	M workers, H inner steps, local batch, accumulation, global/effective batch
runtime	precision, hardware, compiler, gradient checkpointing, distributed backend
evaluation	eval tokens, frequency, downstream harness commit/version
logging	seed, git SHA, wall time, tokens/s, comm bytes, checkpoint paths

10. Ablations Reviewers Will Expect

Ablation	Question answered
Outer Nesterov off	Shows whether DiLoCo is better because of the outer optimizer.
Single-worker SNOO	Directly tests the non-distributed explanation.
H sweep	Tests communication frequency and local drift.

Ablation	Question answered
M sweep	Tests number of replicas independently from H.
Constant global batch	Separates gradient-noise effects from algorithm effects.
Clone workers	Removes useful worker diversity while preserving distributed mechanics.
IID vs non-IID shards	Tests data-distribution sensitivity.
Equal tuning budget	Prevents one baseline from being under-tuned.
Second model scale	Shows the result is not a small-model artifact.
Second corpus	Shows the result is not C4-only.
Communication accounting	Ensures the “low-comm” claim is measured, not assumed.
Seed replication	Determines whether sub-1% loss differences are reliable.

11. Possible Outcomes and How to Write Them

Observed result	Interpretation	Paper angle
SNOO $\approx$ DiLoCo > DP	Most quality gain is optimizer-side; distribution mainly saves communication.	Strong paper: resolves a confound and reframes the free lunch.
DiLoCo > SNOO > DP	Outer optimizer explains part, worker diversity adds a real interaction.	Strongest positive mechanism story; quantify interaction versus H/M.
DiLoCo $\approx$ PeriodicAvg > SNOO/DP	Local trajectories/averaging dominate; Nesterov is less central at your scale.	Interesting contradiction to SNOO-style interpretation; requires careful tuning evidence.
DP $\approx$ all methods	Quality free lunch does not reproduce, but communication savings remain.	Publishable if controls are rigorous and discrepancy with prior setup is diagnosed.
DP > low-comm at matched batch	Earlier advantage was partly a batch/noise artifact.	Strong methodological result; center fairness protocol and negative finding.
Results highly unstable	Outer hyperparameters dominate and claims are brittle.	Paper becomes a robustness/sensitivity study if variance is systematic and well measured.

12. Manuscript Headings and What Each Must Contain

Section	Required content
Abstract	Problem, confound, controlled comparison, one quantitative headline, communication implication. Avoid claiming a free lunch unless both quality and cost evidence support it.
1. Introduction	State the tension: low-communication methods appear to match/beat DDP, while SNOO suggests the outer optimizer itself is unusually strong. List 3-4 contributions.
2. Background	Define DDP AdamW, DiLoCo inner/outer loop, pseudo-gradient, SNOO, PeriodicAvg/FedAvg relation, H and M. Explain fairness pitfalls.
3. Experimental Question: What Must Be Controlled?	Formalize token matching, FLOP accounting, global batch, optimizer tuning budget, and data-order controls. This section is a methodological contribution.

Section	Required content
4. Methods	Describe the factorial variants: DP, SNOO, PeriodicAvg, DiLoCo, Clone-DiLoCo. Give pseudocode or a unified algorithm with switches.
5. Experimental Setup	Models, datasets, tokenizer, hardware, batch sizes, schedules, H/M values, hyperparameter search, evaluation suite, seeds, communication measurement.
6. Reproducing the Free-Lunch Result	DP vs DiLoCo under token/FLOP matching; show loss and communication.
7. Where Does the Gain Come From?	Main causal decomposition: SNOO vs PeriodicAvg vs DiLoCo; recovery fraction and interaction analysis.
8. Batch Size, Worker Diversity, and Local Drift	Global-batch matched runs; clone/IID/skew workers; pseudo-gradient geometry diagnostics.
9. Scaling and Communication Trade-offs	Model-size/H trends and quality-communication Pareto frontier.
10. Discussion	What “free lunch” should mean; when optimizer gains are orthogonal to communication gains; practical recommendations stated as conditional observations, not universal rules.
11. Limitations	Scale, corpus coverage, tuning budget, hardware topology, downstream suite, inability to prove causality beyond interventions tested.
12. Related Work	DiLoCo/Streaming DiLoCo/scaling laws; Lookahead/SNOO/GPA; local SGD/FedAvg/FedOpt/SlowMo; communication-efficient distributed training.
13. Conclusion	One restrained answer to the title question plus open directions.
Appendix A	Full hyperparameter grids and all trials.
Appendix B	Pseudocode and implementation details.
Appendix C	Per-seed curves, failed runs, additional H/M sweeps.
Appendix D	Communication accounting and wall-clock simulation assumptions.
Appendix E	Downstream evaluation details and exact prompts/harness versions.

13. Figures and Tables to Plan Before Training

Item	Content
Fig. 1	Unified diagram: DDP, SNOO, PeriodicAvg, DiLoCo as switches on worker count, local steps, averaging, and outer Nesterov.
Fig. 2	Validation loss vs training tokens for DP and DiLoCo at main scale, with seed bands.
Fig. 3	Main attribution bar/point plot: final loss delta vs DP for SW-AdamW, SNOO, PeriodicAvg, DiLoCo, Clone-DiLoCo.
Fig. 4	H sweep at 2-3 model scales: delta loss vs synchronization interval.
Fig. 5	Worker diversity diagnostics: pseudo-gradient cosine similarity and worker-distance vs training tokens.
Fig. 6	Token-matched vs global-batch-matched comparison.
Fig. 7	Quality vs total communication bytes Pareto frontier.
Table 1	Exact experimental controls and budgets for all methods.
Table 2	Main final metrics with mean/std and communication.

Item	Content
Table 3	Hyperparameter sensitivity and selected configurations.
Table 4	Downstream zero-shot metrics.
Appendix figs	Per-seed curves, second corpus, M sweep, outer LR x momentum heatmaps, communication traces.

14. Project Timeline

Week	Deliverable
Week 1	Literature freeze; reproduce algorithm definitions; implement unified configs; decide model family/tokenizer/corpus.
Week 2	E0 harness parity, checkpoint/resume, exact token/FLOP/communication counters.
Week 3	60M pilot; establish LR and outer-LR/momentum search ranges; debug H and worker sharding.
Week 4	160M single-seed broad sweep: DP, SNOO, PeriodicAvg, DiLoCo across H.
Week 5	Freeze primary hyperparameters and analysis plan; launch 3-seed core E2 comparison.
Week 6	Worker diversity controls: Clone/IID/skew; collect pseudo-gradient geometry metrics.
Week 7	Global-batch matched E4 and communication profiling E6.
Week 8	410M confirmation runs; repeat only predeclared core cells and one H sweep.
Week 9	Second-corpus replication; optional 1B confirmation if budget remains.
Week 10	Statistical analysis, plots, run-audit table, compute/communication accounting.
Week 11	Write full draft, methods first; internal reproducibility review; rerun any broken or ambiguous cell.
Week 12	Polish, appendix, limitations, artifact packaging, anonymous repository snapshot, submission checklist.

15. Go / No-Go Gates

Gate	Pass criterion	Action
Gate 1 - Harness	E0 shows expected DDP parity and exact token accounting.	Do not run expensive sweeps until passed.
Gate 2 - Signal	At 60-160M, at least one of SNOO/DiLoCo differs reproducibly enough from DP to motivate attribution, OR communication savings reproduce cleanly.	If no signal, reduce scope to reproduction/negative result only if methodology is still strong.
Gate 3 - Fairness	Token-matched and batch-matched protocols are both implemented; tuning budgets are documented.	Without this, reviewers can dismiss the core conclusion as confounded.
Gate 4 - Replication	Primary comparison has 3 seeds and one second scale or corpus.	Required before writing a strong general claim.
Gate 5 - Submission	All table values regenerate from logged run summaries; code snapshot reproduces one small run end-to-end.	Artifact and paper must agree.

20. Key References for the Project

1. Charles, Z. et al. (2025). Communication-Efficient Language Model Training Scales Reliably and Robustly: Scaling Laws for DiLoCo. arXiv:2503.09799. [Link](#)
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8. Zhang, M. et al. (2019). Lookahead Optimizer: k steps forward, 1 step back. NeurIPS 2019. [Link](#)